

# Soybean, Tofu, Lentils, and Common Beans in Space

## Executive summary

Soybean is the only one of the three target legumes with meaningful **seed-to-seed spaceflight evidence**. On the ISS, the ADVASC-3 experiment completed a full soybean life cycle in microgravity in about **95 days**, and the returned seeds were reported as healthy, with **germination comparable to commercial terrestrial seed**. On the ground, ESA MELiSSA and related hydroponic studies show that soybeans can germinate reliably, complete a seed-production cycle in about **114–133 days** depending on cultivar, and reach **seed yields on the order of  $\sim 0.6 \text{ kg m}^{-2}$  per crop** in well-managed nitrate-fed NFT systems; NASA biomass-chamber data also place soybean edible dry-mass productivity near  **$6.0 \text{ g m}^{-2} \text{ d}^{-1}$**  under optimized controlled-environment conditions. By contrast, for **lentil** and **common bean**, the strongest space evidence is much thinner: lentil has several classic microgravity studies on **root graviperception, automorphogenesis, and cell-cycle changes**, but not robust edible-yield-to-maturity datasets; common bean has useful hydroponic/CEA studies at 1 g, but I did not locate a comparable seed-to-seed spaceflight record in the primary literature reviewed here. <sup>1</sup>

For **partial gravity**, the evidence base is still mostly indirect. Reviews of altered-gravity plant biology, plus partial-gravity simulation studies, suggest that some root-meristem and tropism phenotypes are substantially perturbed in **microgravity** and remain measurably altered at **lunar gravity ( $\sim 0.16 \text{ g}$ )**, while **Mars gravity ( $\sim 0.38 \text{ g}$ )** may be closer to 1 g for some early root development metrics. However, those Moon/Mars threshold results come mainly from **Arabidopsis**, not legumes. That means the best current conclusion is not that soybean, lentil, or common bean are “proven” at lunar or Martian gravity, but that **soybean is the most technically plausible of the three**, while **legume-specific Moon/Mars gravity validation remains a major research gap**. <sup>2</sup>

Turning soybeans into **tofu in space** is technically plausible, but it is **not a mature space food process**. The terrestrial process is straightforward—soak, grind with water, heat, coagulate, separate curd from whey, press—but every one of those steps becomes harder in microgravity because **phase separation, foam control, liquid containment, gas-liquid disengagement, and drainage-based filtration/pressing are all gravity-dependent on Earth**. Space-adapted tofu production would therefore need a closed, automated unit with **wet milling, heated mixing, capillary or membrane-based separation, positive-pressure pressing, and full wastewater capture**, rather than open vats and cloth filtration. Published tofu-specific mass/power/crew-time budgets for space were not found, so the engineering assessment below necessarily combines primary tofu-processing literature with NASA/ISS fluid-management literature and is partly inferential. <sup>3</sup>

The major system-level risks are not simply “can the plants grow,” but whether a closed food system can stay **sanitary, automatable, resource-efficient, and resilient**. Hydroponics improves nutrient-use efficiency, but recirculating solutions can also spread pathogens if sanitation fails; microgravity can increase plant susceptibility to fungal disease; and foodborne human pathogens persist on ISS surfaces, even

though none have yet been recovered from space-grown lettuce. Root-zone oxygenation, radiation exposure, nutrient recycling from crew wastes, and humidity/light/CO<sub>2</sub> control all remain central design constraints. These risks are manageable in principle, but only with substantial automation, microbial monitoring, and process simplification. 4

The broad conclusion is that **soybean is a credible long-duration candidate crop if the mission truly needs edible seed protein and a soy-based food chain**, but **tofu is probably a second-generation capability**, not a first-wave exploration-food technology. For early lunar and Mars missions, lower-processing options—whole soy foods, soy slurry, roasted seed products, or alternative proteins such as **microalgae and microbial protein**—are likely easier to integrate than a full tofu line. Tofu becomes more attractive only when missions become long enough, closed enough, and large enough that **culinary diversity, calcium fortification, okara reutilization, and loop closure** justify the added hardware and crew/automation burden. 5

## Scope and evidence base

This report prioritizes **peer-reviewed primary papers**, NASA/ESA/JAXA-adjacent technical reviews, NASA technical reports, and conference proceedings. For soybean, the evidence is comparatively rich because it appears in NASA CELSS/BPC work, ESA MELISSA hydroponic screening, and at least one ISS full-life-cycle report. For lentil and common bean, the evidence is much more fragmented: lentil is important in classic space gravitropism experiments, while common bean appears more often in terrestrial hydroponic and nitrogen-fixation studies than in flight programs. 6

Two caveats matter throughout. First, **true legume data in lunar and Martian partial gravity are sparse**; much of the best partial-gravity biology comes from Arabidopsis and other model systems. Second, **no tofu-in-space processing paper** equivalent to the crop papers was found. Where tofu engineering is discussed below, I separate literature-backed facts from explicit engineering inference. 7

## Legume cultivation under microgravity and partial gravity

Soybean has the strongest space-agriculture pedigree of the three species. In ESA MELISSA hydroponic cultivar trials, four soybean cultivars showed **86.9% to 96.8% germination**, a **mean germination time of 4.3 days**, and crop cycles of **114 to 133 days** under controlled-environment NFT. Seed yield averaged about **12 g plant<sup>-1</sup>** across cultivars, and the best cultivar combined shorter stature with high protein content (**35.6% seed protein**). In a separate nitrate-vs-urea NFT study, nitrate-fed soybean reached **604.1 g m<sup>-2</sup> seed yield**, versus only **40.3 g m<sup>-2</sup>** when urea was the sole N source; nitrate-fed plants also consumed **27.2 L plant<sup>-1</sup>** over a **122-day** crop, compared with **6.3 L plant<sup>-1</sup>** under the stunted urea treatment. 8

NASA CELSS/BPC studies support the conclusion that soybean can be productive in controlled environments, but they also show that **nutritional composition is not fixed**. In one NASA soybean composition study across CO<sub>2</sub> regimes, seed protein for cultivars McCall and Pixie was **highest at 1000 ppm CO<sub>2</sub>** and lower at **2000 ppm**, while seed fat peaked at **2000 ppm**. The same report notes that seed ash in the chamber-grown plants was higher than typical field values. A separate paper summarized in search results likewise reported that **leaf fiber and amino-acid/protein patterns changed between 350 and 1000 ppm CO<sub>2</sub>**. In other words, soybean is feasible, but “the same crop” can have materially different food chemistry under space-relevant environmental setpoints. 9

Most importantly, soybean is not just a ground analog candidate. A widely cited review of forty years of space plant growth systems states that **ADVASC-3 on the ISS was the first experiment to grow soybean plants in space**, that a **full life cycle was completed after 95 days**, and that the seeds were healthy with **germination comparable to commercial terrestrial seed**. That does not answer every agronomic question, but it is the most direct evidence in this report that soybean can move through germination, vegetative growth, flowering, pod/seed development, and production of viable next-generation seed in true microgravity. <sup>10</sup>

Lentil sits in a different category: it is a **famous space-biology organism**, but not yet a demonstrated edible-yield crop for space settlement. In the Biorack-era experiments, lentil seedlings grown in microgravity for about **25 hours** still showed strong gravitropic response when later placed under 1 g centrifugation, demonstrating preserved graviperception capacity. Subsequent work reported that lentil roots grown on the ISS under microgravity showed **automorphogenesis** and increased gravisensitivity relative to 1 g-grown roots. Another study found that microgravity altered the **cell cycle in lentil root meristems**, with evidence consistent with a lengthened first cell cycle and/or second G1 phase. These are highly valuable results for gravity biology, but they are measurements of **early seedling/root physiology**, not closed-loop edible production. <sup>11</sup>

For lentil under controlled 1 g hydroponics, a recent 2026 CEA paper is useful as a proxy. It reports that lentil (beluga type) grown in deep-water culture under **200  $\mu\text{mol m}^{-2} \text{s}^{-1}$ , 16 h photoperiod**, and **23/19°C day/night** reached **R5 (early seed)** in about **60 days**, but produced comparatively **low biomass** relative to soybean. The same study highlights that data for non-soy legumes in hydroponics remain scarce. This is encouraging for time-to-harvest, but it is not proof of performance in microgravity, 0.16 g, or 0.38 g. <sup>12</sup>

Common bean is even less mature in the space literature. I did not find a strong seed-to-seed microgravity common-bean paper comparable to soybean ADVASC-3. What exists instead are hydroponic common-bean studies showing that the crop *can* be grown in recirculating systems and that biological N fixation can partly substitute for mineral N under controlled conditions. In a 2023 hydroponic study, a **50–25% mineral-N regime** paired with rhizobial inoculation limited yield loss to about **11.2%** relative to the full-N treatment while saving about **64% of inorganic N** over the crop cycle. The paper also emphasizes that common bean has **poorer BNF capacity than soybean or lentil**, making it more nutritionally attractive than it is agronomically elegant for a highly closed system. <sup>13</sup>

The most defensible crop-by-crop assessment is therefore the following:

| Crop           | Direct spaceflight evidence                                 | Best controlled-environment evidence                                                                 | Growth-stage readiness                                       | Yield evidence most relevant to space                                                                                                    |
|----------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Soybean</b> | Full life cycle on ISS; viable seed returned. <sup>10</sup> | Robust hydroponic/NFT studies in ESA MELISSA; controlled-environment composition data. <sup>14</sup> | Germination through seed production supported. <sup>15</sup> | ~ <b>604 g m<sup>-2</sup></b> seed yield in nitrate NFT; NASA BPC edible DM ~ <b>6.0 g m<sup>-2</sup> d<sup>-1</sup></b> . <sup>16</sup> |

| Crop               | Direct spaceflight evidence                                                            | Best controlled-environment evidence                                              | Growth-stage readiness                                                                                                   | Yield evidence most relevant to space                                                                     |
|--------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Lentil</b>      | Strong early-root microgravity biology; graviperception/cell-cycle work. <sup>17</sup> | Hydroponic CEA indicates ~ <b>60 days to R5</b> but low biomass. <sup>18</sup>    | Early root development supported; edible yield to maturity in altered gravity not demonstrated. <sup>19</sup>            | No robust altered-gravity edible yield per m <sup>2</sup> located. <sup>20</sup>                          |
| <b>Common bean</b> | Weak direct flight evidence located in this review.                                    | Hydroponic common-bean pod production and N-management work at 1 g. <sup>21</sup> | Vegetative and reproductive growth at 1 g hydroponics supported; altered-gravity maturity not established. <sup>21</sup> | Fresh-pod yield reported per plant, not in standardized space-style m <sup>2</sup> metrics. <sup>21</sup> |

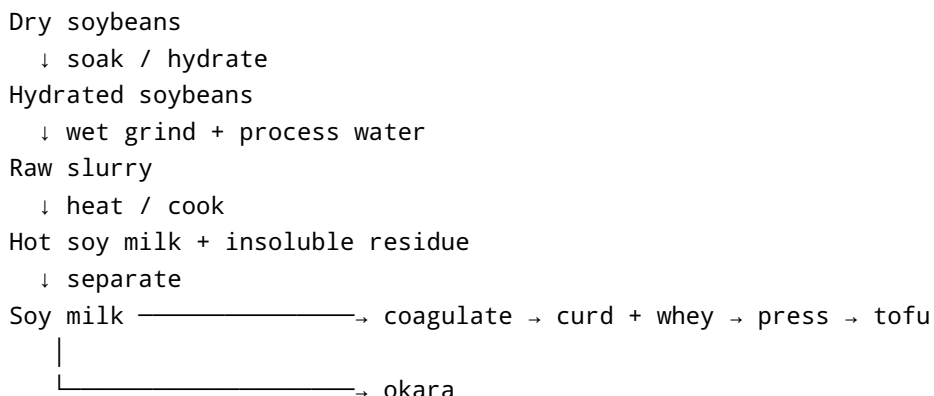
A separate question is **Moon vs Mars gravity**. The best currently available partial-gravity simulations indicate that early root-meristem functions can remain significantly altered at **Moon gravity**, whereas **Mars gravity** may sit closer to terrestrial performance for at least some cell proliferation metrics; one study even suggested the response threshold lies **between Moon and Mars gravity**. Another space experiment found root phototropic behavior emerging in microgravity and being attenuated around **0.1 g–0.3 g**. None of this is legume-specific enough to pick a winner among soybean, lentil, and common bean, but it does strongly imply that **0.16 g and 0.38 g should not be treated as equivalent** for crop design. <sup>22</sup>

## Processing soybeans into tofu in space

On Earth, tofu production follows a compact chain of unit operations: **seed selection and cleaning → soaking → wet grinding with water → heating/simmering soymilk → cooling to coagulant-specific temperature → coagulation → curd/whey separation → pressing → packaged tofu**. In a standardized machine study, soybeans were soaked **12–14 hours**, wet-ground, heated first to about **105°C** and then held near **95°C for 20 minutes**, cooled to the target range for coagulant addition, allowed to coagulate for about **45 minutes**, and then pressed in cloth-lined molds. The same study reports that **2 kg soybean** could produce roughly **10–12 L soymilk** and **2.25–2.40 kg tofu**, while **3 kg soybean** produced **11–13 L soymilk** and **3.30–3.75 kg tofu** depending on solids and coagulant handling. <sup>23</sup>

The likely **minimum space-compatible equipment set** is therefore not exotic, but it is more than a galley appliance. It would include a **sorter/seed dispenser, closed wet mill or colloid grinder, heated reaction vessel, coagulant metering module, closed separator or filter, press/mold module, effluent capture, and sanitation/clean-in-place plumbing**. Small terrestrial tofu machines already integrate grinding, heating, and pressing in one housing; however, the same study also notes operational disadvantages highly relevant to space: **manual soybean feed, manual stone adjustment, manual foam removal, and manual pressing**. In a spacecraft or habitat, those are precisely the tasks that would need to be automated away because current food-system design is heavily constrained by **mass, volume, power, water, preparation equipment, and crew time**. <sup>24</sup>

The easiest way to visualize the process is as a closed material flow:



For mass balance, one soyfoods reference summarized in search results reports that on average about **53% of initial soybean dry mass ends up in tofu**, about **34% in okara**, and about **16% in whey**; because those values come from a secondary source and do not close perfectly to 100%, they should be treated as indicative rather than a strict engineering balance. The key implication is still clear: **okara and whey are not trivial side streams**; they are major streams that must either be reused, reprocessed, or hygienically disposed of. <sup>25</sup>

Coagulant choice matters in space because it changes texture, mineral contribution, and process behavior. The standardized tofu study lists **CaSO<sub>4</sub>, MgSO<sub>4</sub>, MgCl<sub>2</sub>, vinegar, and glucono-δ-lactone (GDL)** among relevant coagulants, with typical application temperatures from about **70°C to 90°C** depending on product style. It also reports that **calcium sulfate gave higher yield than magnesium sulfate**, and that combined CaSO<sub>4</sub> + MgSO<sub>4</sub> conditions gave the highest yields in that machine setup. For space missions, CaSO<sub>4</sub> is attractive because it is simple, stable, and **adds calcium** to the final food; MgCl<sub>2</sub>/nigari is attractive for smooth tofu textures but requires more careful liquid handling because it is more soluble. <sup>26</sup>

From a **gross water-use** standpoint, the same tofu standardization study found that soybeans **roughly doubled in weight after soaking** and used about **2 L of grinding water per kg of initial soybean**. In a spacecraft, the gross process-water burden would be larger once soaking, equipment washdown, and effluent capture are included, but the *net* burden could be moderated if soak water, condensate, and whey-derived water were recoverable through the habitat's water processor. This matters because life-support systems already identify **multi-phase wastewater collection, phase separation, and distillation** as core operational challenges in microgravity. <sup>27</sup>

A simple first-order **thermal estimate** illustrates the energy issue. If the process heats **10 L** of soymilk from 25°C to 100°C, the sensible heat load alone is on the order of **3 MJ** assuming water-like heat capacity; a **45 L** batch would require about **14 MJ** before accounting for inefficiencies, latent losses, or sterilization margins. Those are not impossible loads in a habitat, but they are large enough that tofu is best treated as a **scheduled batch operation** rather than casual meal prep. The case for tofu therefore improves when the mission can exploit waste heat, automation, and closed wash-water recovery. The underlying process temperatures and batch volumes are literature-backed; the MJ values here are engineering calculations derived from those reported process conditions. <sup>23</sup>

The toughest **microgravity-specific obstacles** are phase management and drainage. On Earth, tofu relies heavily on the fact that foam rises, gas disengages, curd settles, whey drains downward, and cloth filtration separates liquid from solids under gravity. NASA's life-support engineering literature repeatedly notes that in microgravity, **gas-liquid phase separation is not buoyancy-driven**, that fluid systems must use special containment strategies, and that root- or process-water systems are vulnerable to **phase-separation and fluid-handling problems**. Boiling and condensation are also altered; stable boiling is possible in microgravity under some conditions, but the thermal/fluid behavior is different enough that process vessels cannot simply be “kitchen hardware in space.” That means a practical tofu unit would need **capillary separators, membranes, screw or piston pressing, or small centrifugation steps**, not open settling or free drainage. <sup>28</sup>

For **crew time**, the literature gives a clear warning even when it does not quantify tofu-specific minutes. The Biosphere 2 food system—an imperfect but still relevant analog—spent about **45% of crew time** on agriculture and food-related work, including food processing and cooking. NASA food-system reviews likewise emphasize that long-duration mission design must reduce reliance on crew labor. Since the standardized tofu machine still required manual seed feed, foam management, and pressing, a direct terrestrial-to-space translation would be unacceptable; a viable space tofu process would need near-total automation or integration into a broader smart galley/bioprocessing module. <sup>29</sup>

A concise engineering view of the process is below.

| Process step         | Terrestrial condition from cited studies                                                    | Space-specific issue                                             | Most plausible mitigation                                                                        |
|----------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Soaking/hydration    | ~12–14 h soak; beans roughly double in weight. <sup>23</sup>                                | Free liquid containment; contamination risk during long wet hold | Closed soak cartridges, UV/thermal sanitation, timed drain-to-recovery                           |
| Wet grinding         | ~2 L grinding water per kg soybean. <sup>23</sup>                                           | Slurry transport, bubble entrainment, cleaning burden            | Closed wet mill with positive-displacement pumping and CIP                                       |
| Heating/cooking      | ~95–105°C process range; minimum ~10 L in cited machine. <sup>23</sup>                      | Foam removal, altered boiling/condensation, thermal control      | Jacketed vessel, controlled nucleation, antifoam-compatible design, condenser/capillary gas trap |
| Coagulation          | CaSO <sub>4</sub> , MgSO <sub>4</sub> , MgCl <sub>2</sub> , GDL all relevant. <sup>30</sup> | Uniform mixing without shear damage; bubble-free dosing          | Metered injection and low-shear recirculation loop                                               |
| Curd/whey separation | Cloth-lined box and gravity-assisted handling on Earth. <sup>23</sup>                       | Gravity separation fails; whey does not drain “down”             | Membrane separation, screw press, piston press, or centrifuge                                    |
| Pressing/forming     | Manual pressing in cited machine. <sup>23</sup>                                             | Crew labor and fluid leakage                                     | Sealed compression mold with captured whey return                                                |

| Process step   | Terrestrial condition from cited studies                         | Space-specific issue                         | Most plausible mitigation                                                                      |
|----------------|------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------|
| Waste handling | Okara byproduct and whey/press liquor substantial. <sup>31</sup> | Shelf life, microbial growth, water recovery | Integrate okara into breads/extrudates; route whey/wash water to WPA or BLSS nutrient recovery |

## Limitations and failure modes

The most immediate biological limitation is **disease ecology**, not just plant growth rate. Spaceflight and modeled microgravity are associated with altered pathogen behavior and altered plant susceptibility. A soybean spaceflight study found that seedlings grown in microgravity were **more susceptible to colonization by the fungal root-rot pathogen *Phytophthora sojae***. A broader 2023 review likewise concludes that microgravity tends to increase plant susceptibility to fungal and oomycete phytopathogens and alters relationships with symbionts. That makes legume systems especially sensitive because they are often discussed together with **rhizobial symbiosis**, but in space the line between “useful microbe” and “management headache” is thin. <sup>32</sup>

Food safety adds another layer. NASA- and ISS-related reviews note that although **human pathogens have not yet been recovered from space-grown produce**, genera that include human pathogens have been found colonizing space-grown plant tissues, and multiple foodborne pathogenic genera are known to persist on ISS surfaces. For any soy-to-tofu chain, this is not just a greenhouse problem; it is a **bioprocess sanitation problem**, because wet grinding, heating, holding, separation, and reuse of water all create classic food-microbiology critical control points. <sup>33</sup>

Root-zone physics is the core engineering limitation for all three legumes. NASA’s “Farming in Space” and root-zone workshops emphasize that in microgravity designers must deliver **water, nutrients, and oxygen** simultaneously while preventing hypoxia and maintaining plant support. Hydroponic/aeroponic systems reduce substrate mass and may scale better, but they are also more exposed to **gas-liquid phase-separation failures**. Conversely, substrate-based systems offer passive water buffering but can suffer from poor aeration when evapotranspiration is low and humidity is high. This trade is one reason many early flight units used solid media, while newer systems like **XROOTS** are explicitly testing hydroponic and aeroponic nutrient delivery for larger-scale future production. <sup>34</sup>

Closed-loop nutrient recycling is promising, but it is not plug-and-play. Hydroponic recirculation improves water and nutrient efficiency, and soilless systems can minimize fertilizer loss, but **using recovered nutrients from waste streams** remains a major unresolved issue. A recent review framed around ESA MELISSA notes that much work remains before nutrients recovered from urine and other organic wastes can be dependably used for higher plants in hydroponics. NASA also continues to investigate recovery of nutrients from **inedible plant biomass** by water leaching and bioreactor processing. For legumes, this matters twice: once for the crop itself, and again for the large side streams from tofu making, especially okara and whey. <sup>35</sup>

Radiation is another serious uncertainty. Reviews of exploration-class space radiation note that plant responses to ionizing radiation include **DNA damage, mutation, growth and reproductive effects**, but also that direct crop-relevant evidence remains limited compared with microgravity studies. In other words,

gravity biology is ahead of radiation agronomy. That matters especially for lunar greenhouses and Mars transit systems, because a soybean genotype that looks excellent in Earth chambers may not behave the same way under chronic mixed-field radiation. <sup>36</sup>

Finally, there are crop-specific agronomic limitations. Soybean's strengths are its documented seed production and dense protein/lipid profile, but its weaknesses are **long cycle time**, large non-edible biomass, and a need for reasonably high light and careful N management. Lentil may offer a **shorter path to seed-stage development** under CEA, but evidence for altered-gravity yield is weak. Common bean can work hydroponically, but its **BNF efficiency is poorer** than soybean or lentil, reducing one of the main reasons to prefer legumes in a closed system. <sup>37</sup>

## Comparison with other crops and alternative proteins

If the mission objective is simply “maximize protein resilience per unit system burden,” soybeans must be compared not only with lentils and common beans, but also with **other higher plants** and **non-plant protein platforms**. NASA crop-productivity summaries show that under controlled conditions, **wheat, potato, and tomato** all achieved higher edible dry-mass productivity than soybean in the biomass chamber, though of course with very different nutritional profiles and processing requirements. Soybean's edible dry-mass productivity near **6.0 g m<sup>-2</sup> d<sup>-1</sup>** is respectable, but it is not dominant in edible biomass terms; its advantage is that much of that edible dry mass is **protein- and oil-rich seed**. <sup>38</sup>

Microalgae and microbial protein are attractive because they can decouple protein production from flowering, seed set, and large shoot systems. A review on photobioreactors for BLSS concludes that microbial photosynthesis is a strong fit for regenerative life support. For **Arthrospira/“spirulina”**, terrestrial production references report mean annual dry-biomass productivities around **7.35–8.14 g m<sup>-2</sup> d<sup>-1</sup>** in suitable systems, while an industrial LCA reported about **54.48 kWh electricity** and **8.360 m<sup>3</sup> freshwater per kg edible wet biomass** in one geothermal site workflow. Those industrial figures are not directly transferable to a closed habitat—they reflect an Earth production chain, not a space-optimized loop—but they illustrate the basic trade: algae can be highly productive and continuous, yet **gas exchange, dewatering, contamination control, and energy use** become the central engineering problems. <sup>39</sup>

Hydrogen-oxidizing **single-cell protein** pushes that logic further. A recent space-oriented analysis of HOB-based SCP estimates total electricity consumption of roughly **26.8–37.5 kWh kg<sup>-1</sup> SCP**, arguing that such systems can make food in space without agricultural land in the usual sense. That is a very different design philosophy from crop production: little or no canopy, no pollination risk, rapid production, and potentially very high areal efficiency if power is abundant. The drawback is equally clear: this is a **power-first food technology**, and the resulting food product is a microbial ingredient that usually requires downstream formulation rather than direct culinary familiarity. <sup>40</sup>

Insects are viable enough to be on ESA's radar, but they remain more speculative as an exploration food platform than algae or microbial protein. ESA notes that worms and other bugs are **high in protein, fatty acids, iron, zinc, and B vitamins**, and mealworms/crickets are prominent candidates. The strongest arguments for insects are **small breeding space, high nutrient density, and capacity to valorize side streams**. The weakest part of the case is that **space-integrated quantitative resource budgets are still immature**, and public datasets remain much thinner than for crops or phototrophic microbes. <sup>41</sup>

The comparison below is therefore intentionally conservative. Where the literature supports a numeric value, one is shown; where it does not, the field is marked as not yet standardized.

| Candidate                      | Time to harvest or productive state                                                                  | Quantitative metric available in reviewed sources                                                                                         | Main systems advantage                                                    | Main systems drawback                                                              |
|--------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Soybean</b>                 | ~95 d in ISS ADVASC-3; ~114–133 d in hydroponic cultivar trials; 122 d in nitrate NFT. <sup>42</sup> | Seed yield <b>604.1 g m<sup>-2</sup></b> in nitrate NFT; edible DM <b>~6.0 g m<sup>-2</sup> d<sup>-1</sup></b> in NASA BPC. <sup>16</sup> | Proven seed-to-seed in space; dense protein/oil seed                      | Long cycle; significant non-edible biomass; more processing needed if tofu desired |
| <b>Lentil</b>                  | ~60 d to R5 in 1 g hydroponic CEA. <sup>18</sup>                                                     | No robust altered-gravity yield per m <sup>2</sup> located.                                                                               | Shorter development than soybean; strong legume nutrition appeal          | Limited spaceflight agronomy evidence beyond roots                                 |
| <b>Common bean</b>             | Fresh-pod hydroponic production demonstrated at 1 g. <sup>21</sup>                                   | Pod yield in cited work reported per plant, not standardized space m <sup>2</sup> . <sup>21</sup>                                         | Familiar food; hydroponics already workable                               | Weak direct flight evidence; lower BNF efficiency than soybean/lentil              |
| <b>Wheat</b>                   | Established space-crop candidate. <sup>38</sup>                                                      | Edible DM <b>~11.3 g m<sup>-2</sup> d<sup>-1</sup></b> in NASA BPC summary. <sup>38</sup>                                                 | Very well studied; bread/pasta versatility                                | Lower protein density than soybean; milling/baking still needed                    |
| <b>Potato</b>                  | Established BLSS crop. <sup>38</sup>                                                                 | Edible DM <b>~18.4 g m<sup>-2</sup> d<sup>-1</sup></b> in NASA BPC summary. <sup>38</sup>                                                 | Very high edible biomass productivity                                     | Not a protein-dense crop                                                           |
| <b>Arthrospira / spirulina</b> | Continuous culture rather than harvest cycle. <sup>43</sup>                                          | <b>~7.35–8.14 g dry biomass m<sup>-2</sup> d<sup>-1</sup></b> in cited terrestrial systems. <sup>44</sup>                                 | Continuous, compact, loop-friendly                                        | Dewatering, contamination, gas management, acceptance                              |
| <b>HOB single-cell protein</b> | Continuous bioprocess. <sup>45</sup>                                                                 | <b>26.8–37.5 kWh kg<sup>-1</sup> SCP</b> in cited analysis. <sup>45</sup>                                                                 | No field-like canopy; potentially high protein productivity per footprint | Power-intensive; formulation and acceptability challenges                          |

| Candidate                 | Time to harvest or productive state                                   | Quantitative metric available in reviewed sources                     | Main systems advantage                                                 | Main systems drawback                                                   |
|---------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Mealworms/ insects</b> | Rapid life cycles relative to many vertebrate proteins. <sup>46</sup> | Space-relevant quantitative budgets still sparse in reviewed sources. | High nutrient density; small breeding space; can valorize side streams | Immature space systems integration; acceptability and hygiene questions |

For mission architecture, that comparison leads to a pragmatic hierarchy. If the mission needs **fresh psychologically meaningful crops**, soybean can justify itself. If the mission needs **maximum edible biomass**, potato and wheat remain formidable. If the mission needs **compact protein factories, algae and microbial protein** may outperform seed crops once sufficient power and bioprocessing reliability are available. If the mission needs **culinary familiarity plus protein**, soy may still be the best bridge technology—especially before crew acceptability catches up with microbial and insect foods. <sup>47</sup>

## Engineering mitigation strategies and research gaps

The first engineering rule is **simplify the food chain before optimizing it**. For soy, that means flying the crop first as **seed or minimally processed whole-soy foods**, not as a full tofu line. A habitat can derive nutritional value from roasted soy, soy pastes, sprouted soy foods, or soy beverage intermediates with fewer unit operations than tofu. Only after stable soybean growth, sanitation, loop closure, and gas-liquid handling are demonstrated does tofu become a rational add-on. This is consistent with NASA food-system thinking, which treats mass, volume, water, power, preparation equipment, and crew time as strong limiting variables. <sup>48</sup>

The second rule is **design the crop-production hardware and the food-processing hardware together**. NASA plant-system literature repeatedly shows that root-zone hydration, gas management, and containment are coupled problems; tofu processing has the same coupling in another form. A mission that grows soybeans hydroponically or aeroponically should assume from the start that post-harvest soy processing will need the same capillary-fluid, separator, pump, vent, and sanitation competencies already being developed for space agriculture and water recovery. The right architecture is not “greenhouse plus kitchen,” but an integrated **bioresource module** with shared fluidics, sterilization, monitoring, and waste routing. <sup>49</sup>

The third rule is **prefer process routes that minimize separation steps**. In tofu terms, the more the process depends on free settling, cloth drainage, and manual pressing, the worse it fits microgravity. Space-compatible research should therefore prioritize: membrane-assisted soy milk clarification; capillary-managed or low-g centrifuge separation; **coagulation in the final container** where possible; and reusable process cartridges that can be sanitized in place. These are engineering inferences from the reviewed fluid-physics and tofu-process literature, but they directly target the known failure modes of phase separation and drainage in microgravity. <sup>50</sup>

The fourth rule is **treat microbes as both enablers and hazards**. Legumes are attractive partly because of nitrogen fixation, yet common bean data already show that BNF can be inconsistent even at 1 g

hydroponics, while soybean hydroponic inoculation has not consistently improved performance. At the same time, microgravity can increase pathogen susceptibility and alter microbial communities. The right mitigation is not “eliminate microbes,” but rather tightly engineered microbiology: defined inocula, continuous water-loop sanitation, biosensors, and a systems-level “One Health” view linking crew, crops, and habitat microbiomes. <sup>51</sup>

The most important **research gaps** are now very clear.

| Gap                                                                          | Why it matters                                                                     | Most useful next experiment                                                                                                             |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Legume-specific partial gravity data</b> for soybean, lentil, common bean | Moon and Mars cannot be inferred confidently from Arabidopsis alone. <sup>52</sup> | Multi-level centrifuge experiment with all three crops from germination to at least pod/seed set at $\mu\text{g}$ , 0.16 g, 0.38 g, 1 g |
| <b>Space-valid yield and nutrient composition</b> for lentil and common bean | Current evidence is mostly root biology or 1 g hydroponics. <sup>53</sup>          | Long-duration CEA trials under space-relevant $\text{CO}_2$ /light/humidity, then flight validation                                     |
| <b>Tofu process demonstrator in reduced gravity</b>                          | The crop may work while the food process fails.                                    | Bench-scale capillary/membrane tofu unit on clinostat/parabolic flight/ISS tech demo                                                    |
| <b>Okara and whey valorization</b> in BLSS                                   | Tofu is only attractive if side streams are reused. <sup>54</sup>                  | Integrate okara into breads/extrudates and route whey to nutrient/water recovery                                                        |
| <b>Microbial control in wet soy processing</b>                               | Wet grinding/coagulation is a food-safety hotspot. <sup>33</sup>                   | HACCP-style spacefood study with pathogen challenge surrogates and CIP validation                                                       |
| <b>Radiation-by-gravity interactions</b> in seed legumes                     | Current agronomy is gravity-heavy and radiation-light. <sup>55</sup>               | Chronic low-dose mixed-radiation plus altered-gravity trials for seed set and seed quality                                              |

The bottom line for mission planners is straightforward. If the target is a **near-term lunar or Mars mission**, soybean should be treated as a **mid-complexity crop** and tofu as a **high-complexity processed food**. If the target is a **long-stay settlement with stronger closure requirements**, then soybean’s ability to deliver protein-rich seeds and multiple culinary endpoints becomes much more valuable, and tofu becomes worth prototyping—provided the mission is also willing to invest in advanced multiphase fluid handling, sanitation, automation, and side-stream reuse. <sup>56</sup>

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